

Colby Crutcher

Secure Coding

Lab 7

Challenge 1:

```
colby@Colby-PC: /mnt/c/Users/wcrut/OneDrive/Documents/GitHub/SecureCoding/Labs/Lab7$ file challenge1
challenge1: ELF 32-bit LSB executable, Intel 80386, version 1 (SYSV), dynamically linked, interpreter /lib/ld-linux.so.2, BuildID[sha1]=062fd801b3b444d70eb9698e0e0983da65232d56, for GNU/Linux 3.2.0, not stripped
```

Figure 1: File Endianness

```
pwndbg> checksec
File:      /home/ccrutter/lab7/challenge1
Arch:      i386
RELRO:     Partial RELRO
Stack:     No canary found
NX:        NX enabled
PIE:       No PIE (0x8048000)
Stripped:  No
pwndbg> 
```

Figure 2: checksec

- Arriving at the EIP value:

```
Skipped. NO
pwndbg> disas main
Dump of assembler code for function main:
    0x080491ab <+0>:    push    ebp
    0x080491ac <+1>:    mov     ebp,esp
    0x080491ae <+3>:    and     esp,0xffffffff
    0x080491b1 <+6>:    call    0x8049204 <__x86.get_pc_thunk.ax>
    0x080491b6 <+11>:   add     eax,0x2e4a
    0x080491bb <+16>:   call    0x8049172 <vuln>
    0x080491c0 <+21>:   mov     eax,0x0
    0x080491c5 <+26>:   leave
    0x080491c6 <+27>:   ret
End of assembler dump.
pwndbg> █
```

Figure 3: Main Disassembled